

0.1 Eigenvalues, Eigenvectors, and the Spectral Theorem for Symmetric/Hermitian Matrices

To understand the intrinsic geometric behavior of a linear operator $T : V \rightarrow V$, we isolate the specific directions in V that are strictly scaled by T , invariant in their span. We assume V is a finite-dimensional inner product space over the field F (where F is either R or C), equipped with an inner product $\langle \cdot, \cdot \rangle$.

Let $T : V \rightarrow V$ be a linear operator. A scalar $\lambda \in F$ is an **eigenvalue** of T if there exists a non-zero vector $v \in V$ such that $T(v) = \lambda v$. The vector v is called an **eigenvector** corresponding to λ .

The **adjoint** of a linear operator $T \in \mathcal{L}(V)$, denoted T^* , is the unique linear operator on V such that $\langle T(u), v \rangle = \langle u, T^*(v) \rangle$ for all $u, v \in V$. An operator T is **self-adjoint** (or Hermitian) if $T = T^*$. In the context of real matrix representations, this is equivalent to a symmetric matrix ($A = A^T$).

The structure of self-adjoint operators is remarkably rigid. We first prove two foundational lemmas regarding their spectra.

[Real Spectrum of Self-Adjoint Operators] If $T : V \rightarrow V$ is a self-adjoint operator, then all eigenvalues of T are strictly real.

Let $\lambda \in C$ be an eigenvalue of T , and let $v \in V$ be a corresponding non-zero eigenvector. We evaluate the inner product $\langle T(v), v \rangle$ in two distinct ways. On one hand, applying the eigenvalue definition directly yields:

$$\langle T(v), v \rangle = \langle \lambda v, v \rangle = \lambda \langle v, v \rangle$$

On the other hand, utilizing the self-adjoint property $T = T^*$:

$$\langle T(v), v \rangle = \langle v, T^*(v) \rangle = \langle v, T(v) \rangle = \langle v, \lambda v \rangle = \bar{\lambda} \langle v, v \rangle$$

Equating the two results gives $\lambda \langle v, v \rangle = \bar{\lambda} \langle v, v \rangle$. Since v is a non-zero eigenvector, $\langle v, v \rangle = \|v\|^2 \neq 0$. Dividing by $\|v\|^2$ yields $\lambda = \bar{\lambda}$, proving that $\lambda \in R$.

[Orthogonality of Eigenspaces] Let $T : V \rightarrow V$ be a self-adjoint operator. If λ_1 and λ_2 are distinct eigenvalues of T , then their corresponding eigenvectors v_1 and v_2 are orthogonal.

Suppose $T(v_1) = \lambda_1 v_1$ and $T(v_2) = \lambda_2 v_2$, with $\lambda_1 \neq \lambda_2$. We evaluate the inner product $\langle T(v_1), v_2 \rangle$:

$$\langle T(v_1), v_2 \rangle = \langle \lambda_1 v_1, v_2 \rangle = \lambda_1 \langle v_1, v_2 \rangle$$

Alternatively, utilizing the adjoint property and the fact that eigenvalues of T are real (hence $\bar{\lambda}_2 = \lambda_2$):

$$\langle T(v_1), v_2 \rangle = \langle v_1, T^*(v_2) \rangle = \langle v_1, T(v_2) \rangle = \langle v_1, \lambda_2 v_2 \rangle = \lambda_2 \langle v_1, v_2 \rangle$$

Equating these expansions gives $\lambda_1 \langle v_1, v_2 \rangle = \lambda_2 \langle v_1, v_2 \rangle$, which rearranges to:

$$(\lambda_1 - \lambda_2) \langle v_1, v_2 \rangle = 0$$

Because $\lambda_1 \neq \lambda_2$, we must have $\langle v_1, v_2 \rangle = 0$, proving that v_1 and v_2 are orthogonal.

We are now equipped to prove the central result of finite-dimensional spectral theory, which demonstrates that self-adjoint operators can be completely decoupled into independent orthogonal 1-dimensional projections.

[Spectral Theorem for Finite-Dimensional Spaces] Let V be a finite-dimensional inner product space over F , and let $T \in \mathcal{L}(V)$ be a self-adjoint operator. Then V admits an orthonormal basis consisting entirely of eigenvectors of T .

We proceed by mathematical induction on $n = \dim(V)$.

Base Case: If $n = 1$, any non-zero vector $v \in V$ is an eigenvector of T . Normalizing v by setting $e_1 = \frac{v}{\|v\|}$ provides the required orthonormal basis. The theorem trivially holds.

Inductive Step: Assume the theorem is true for all inner product spaces of dimension $n - 1$. Let $\dim(V) = n$. By the Fundamental Theorem of Algebra, the characteristic polynomial of T possesses at least one root over C . Because T is self-adjoint, Lemma 3.2.1 guarantees this root is real, ensuring T has at least one real eigenvalue λ_1 and a corresponding non-zero eigenvector v_1 . We normalize this vector to obtain a unit eigenvector $e_1 = \frac{v_1}{\|v_1\|}$.

Define $W = \text{span}(e_1)^\perp = \{u \in V \mid \langle u, e_1 \rangle = 0\}$. Since $\dim(V) = n$ and $\dim(\text{span}(e_1)) = 1$, standard properties of orthogonal complements dictate that $\dim(W) = n - 1$.

We claim that W is invariant under T (i.e., $T(W) \subset W$). Let $w \in W$. We must show that $T(w) \in W$, which requires $\langle T(w), e_1 \rangle = 0$. Using the self-adjointness of T and the fact that $T(e_1) = \lambda_1 e_1$:

$$\langle T(w), e_1 \rangle = \langle w, T^*(e_1) \rangle = \langle w, T(e_1) \rangle = \langle w, \lambda_1 e_1 \rangle = \lambda_1 \langle w, e_1 \rangle$$

Because $w \in W = \text{span}(e_1)^\perp$, $\langle w, e_1 \rangle = 0$. Thus, $\langle T(w), e_1 \rangle = 0$, proving $T(w) \in W$. Therefore, the restriction of T to W , denoted $T|_W : W \rightarrow W$, is a well-defined linear operator.

Furthermore, $T|_W$ inherits self-adjointness from T . For any $x, y \in W$, $\langle T|_W(x), y \rangle = \langle T(x), y \rangle = \langle x, T(y) \rangle = \langle x, T|_W(y) \rangle$.

Because $T|_W$ is a self-adjoint operator on the $(n - 1)$ -dimensional inner product space W , the inductive hypothesis applies. There exists an orthonormal basis $\{e_2, e_3, \dots, e_n\}$ for W consisting of eigenvectors of $T|_W$ (and consequently, eigenvectors of T).

We construct the set $\mathcal{B} = \{e_1, e_2, \dots, e_n\}$. Since e_1 is orthogonal to every vector in W , and $\{e_2, \dots, e_n\}$ is an orthonormal set in W , $\mathcal{B}$ forms an orthonormal set of n vectors in the n -dimensional space V . Thus, $\mathcal{B}$ is an orthonormal basis for V composed entirely of eigenvectors of T , completing the induction.